

Basic Principles of Medicine 2

Module: Digestion and Metabolism

Lecture: DM 28

Digestion and Absorption

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Acknowledgement

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Objectives: DLA: Digestion and Absorption

- Before DM Lecture 28

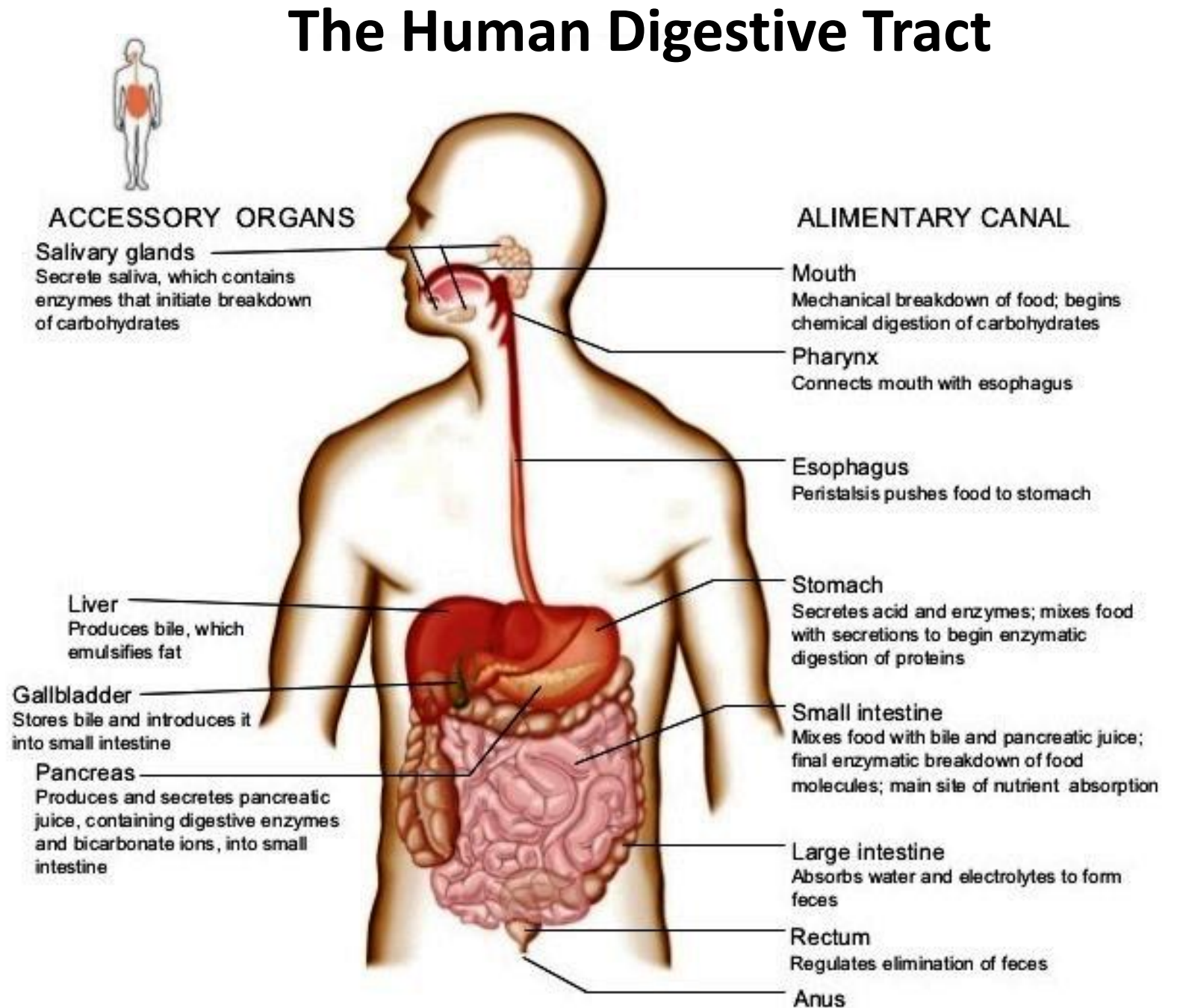
SOM. MK.I.BPM2.3.DM.3.BCHM.1237	Describe digestion in mouth and stomach (role of salivary amylase and gastric lipase).
SOM. MK.I.BPM2.3.DM.3.BCHM.1238	Describe role of gastrin and activation of pepsinogen in stomach. Discuss consequences of insufficient acid production in stomach and the effect of antacids. Discuss gastric digestion of milk TAGs.
SOM. MK.I.BPM2.3.DM.3.BCHM.1239	Explain mechanisms involved in release of cholecystokinin and secretin. Review functions of cholecystokinin and secretin. Indicate role of HCO_3^- in pancreatic secretion and bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1240	Outline digestion and absorption of carbohydrates. Describe reactions of disaccharidases in brush border of intestinal cells.
SOM. MK.I.BPM2.3.DM.3.BCHM.1241	Describe lactose intolerance and differentiate primary, secondary and congenital lactose intolerance based on age of onset and primary defect.
SOM. MK.I.BPM2.3.DM.3.BCHM.1242	Outline lipid digestion and absorption and describe the significance of bile.
SOM. MK.I.BPM2.3.DM.3.BCHM.1243	Discuss differences between primary and secondary bile acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1244	Outline role of liver in synthesis of primary bile acids from cholesterol. Review formation of bile salts and their release into bile.

Lecture Objectives: Digestion and Absorption

SOM. MK.I.BPM2.3.DM.3.BCHM.1245	Outline digestion and absorption of dietary carbohydrates, proteins, lipids and nucleic acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1246	Outline digestion of dietary proteins and absorption of amino acids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1247	Describe the proteolytic activation of zymogens using trypsinogen. Identify the specific cleavage sites of digestive proteases like trypsin, chymotrypsin and elastase.
SOM. MK.I.BPM2.3.DM.3.BCHM.1248	Describe absorption of monosaccharides into intestinal mucosal cells.
SOM. MK.I.BPM2.3.DM.3.BCHM.1249	Describe reaction catalyzed by pancreatic lipase and review importance of bile salts and colipase. Highlight their role in emulsification of dietary lipids.
SOM. MK.I.BPM2.3.DM.3.BCHM.1250	Discuss the MAG pathway of TAG formation in mucosal cells. Outline formation and fate of chylomicrons.
SOM. MK.I.BPM2.3.DM.3.BCHM.1251	Describe enterohepatic circulation of bile salts. Summarize functions of bile salts.
SOM. MK.I.BPM2.3.DM.3.BCHM.1252	Indicate mechanisms in various causes of steatorrhea (pancreatic disease, biliary tract obstruction, intestinal mucosal disease).
SOM. MK.I.BPM2.3.DM.3.BCHM.1253	Discuss bile composition and factors facilitating gallstone formation (cholelithiasis) and predict effect of bile duct obstruction on lipid digestion and absorption.
SOM. MK.I.BPM2.3.DM.3.BCHM.1254	Predict consequences of pancreatic disease (cystic fibrosis and pancreatitis) on digestion and discuss basis for dietary management.

Recommended Reading

The human digestive tract (alimentary canal), including the major structures and accessory organs.

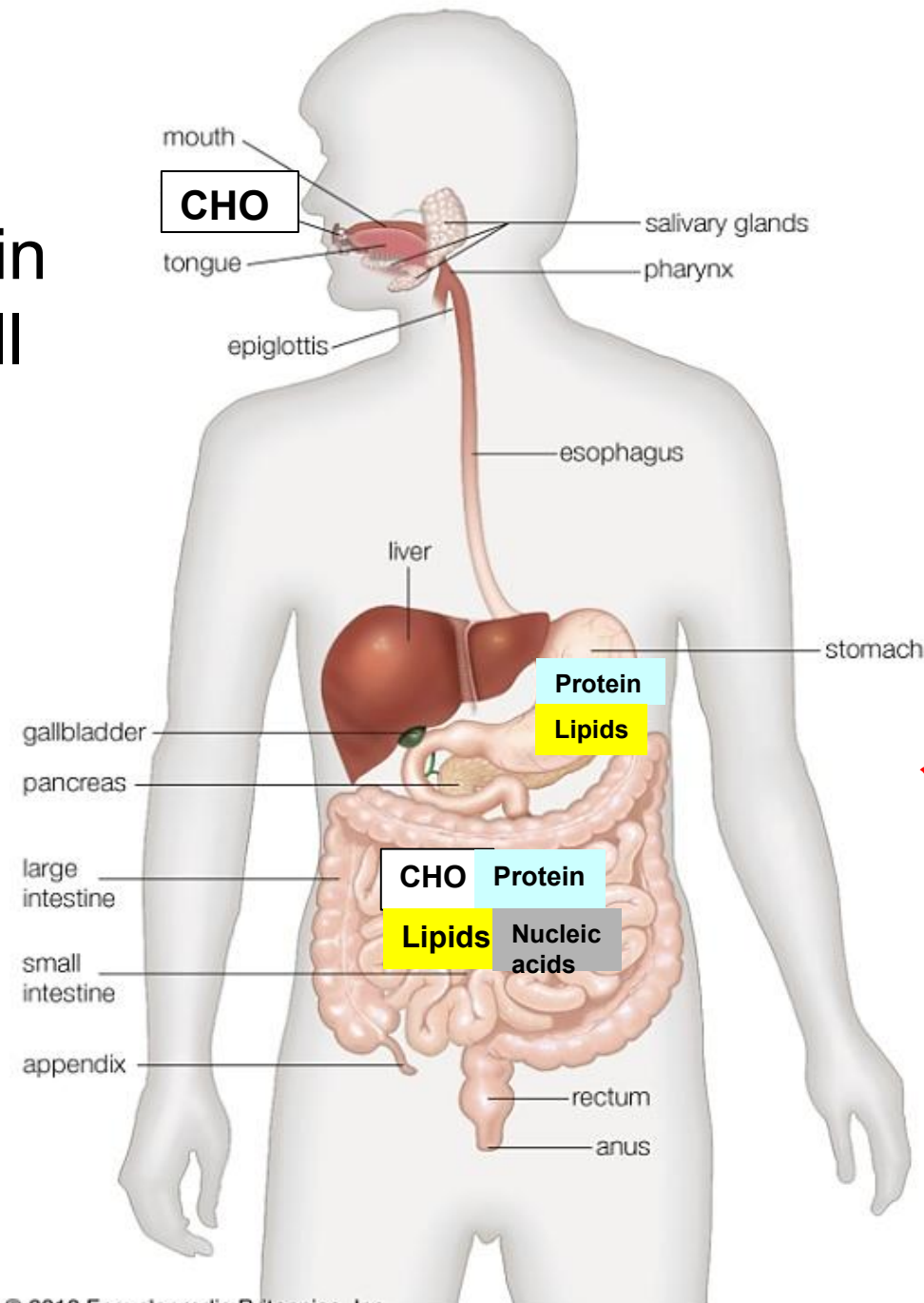


Recommended Reading

- Lippincott Illustrated Reviews (**Lippincott**) 8th edition:
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - **Dietary Protein Digestion:** [19. Amino Acids: Nitrogen Disposal | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - **Dietary Carbohydrate Digestion:** [Introduction to Carbohydrates | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - **Lipid Digestion:** [15. Dietary Lipid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - **Degradation of Dietary Nucleic Acids:** [Nucleotide Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library
- Review the DLA video related to this lecture before class.

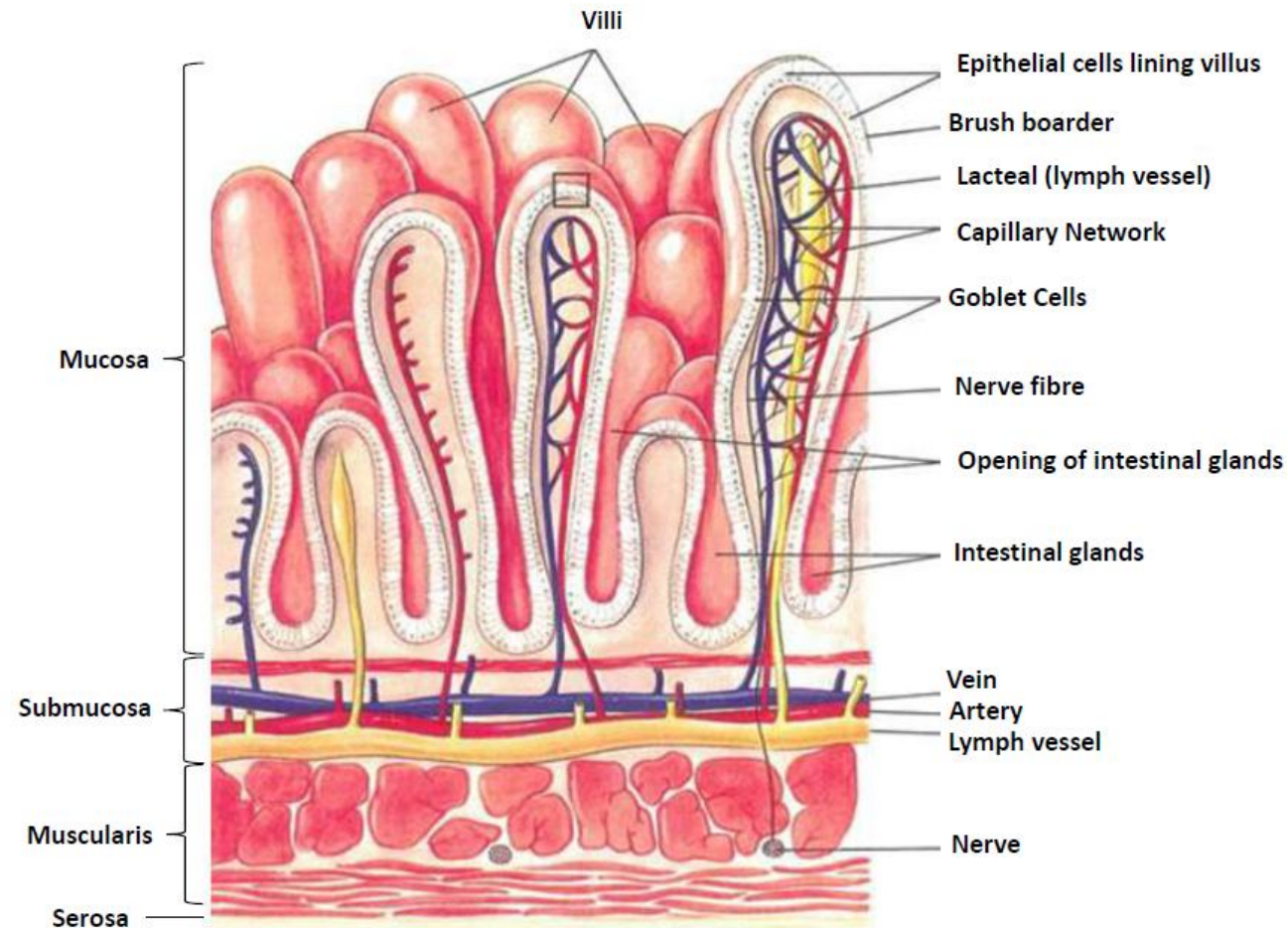
Digestion

- ✓ Digestion starts in the mouth and continues in the stomach and small intestine.



- ✓ **Nutrient absorption begins in duodenum.**

Absorption of all macronutrients takes place in the intestinal mucosa cells.



Fate of Absorbed Macronutrients

- The amino acids, monosaccharides, ribose, deoxyribose and pyrimidines are released into the hepatic portal vein and first reach the liver.
- Purines are degraded in the intestinal mucosa cells to uric acid, enters the bloodstream and eventually excreted in the urine via the kidney.
- Lipids are released into the lymph inside of chylomicrons which later join the blood circulation.

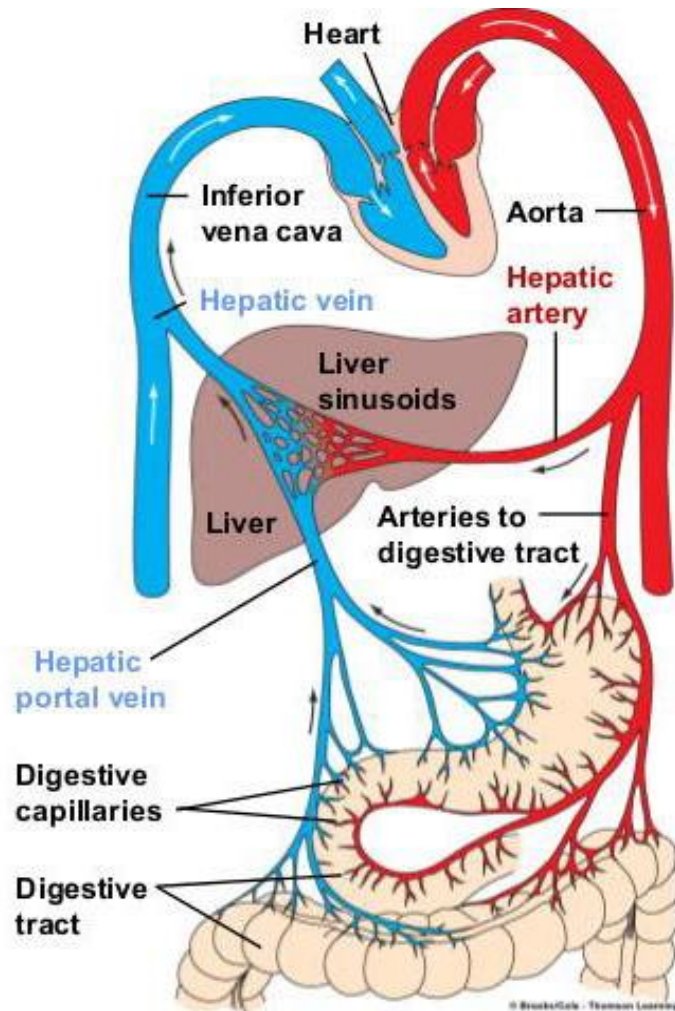
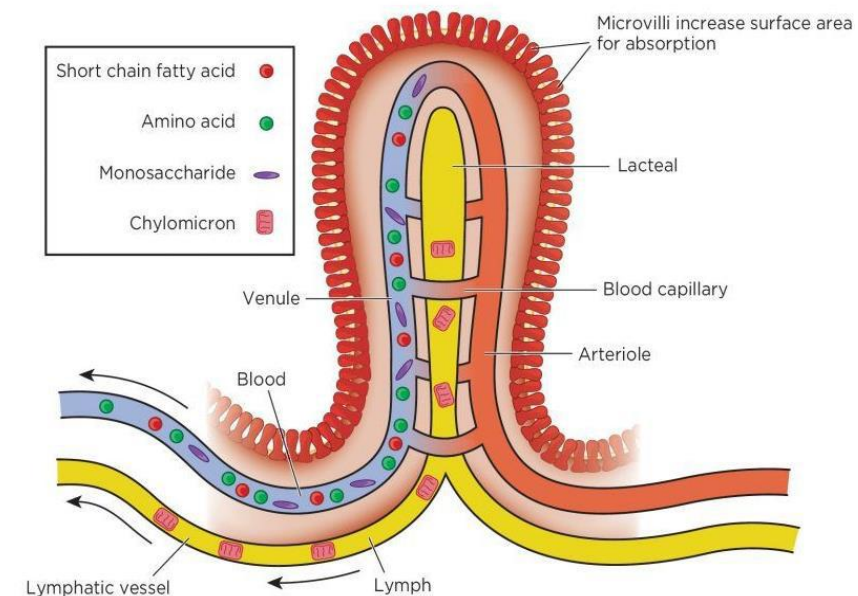
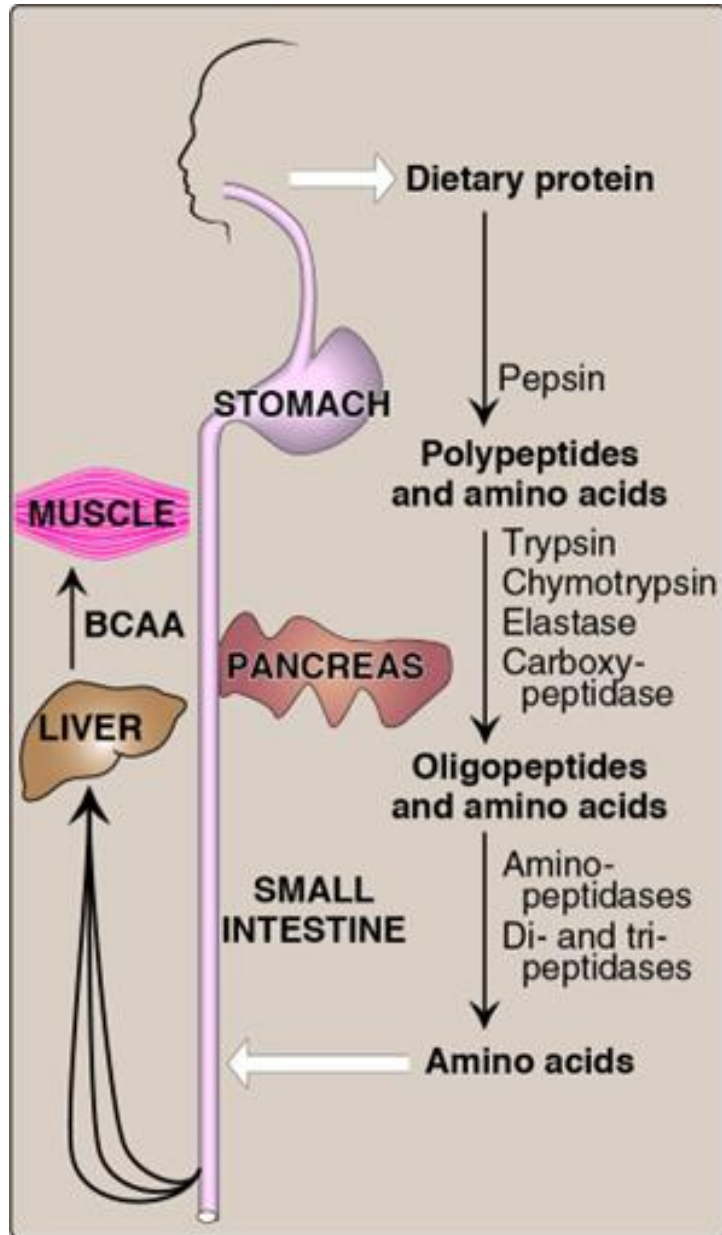


Illustration of the hepatic portal system



BIG picture: Protein Digestion

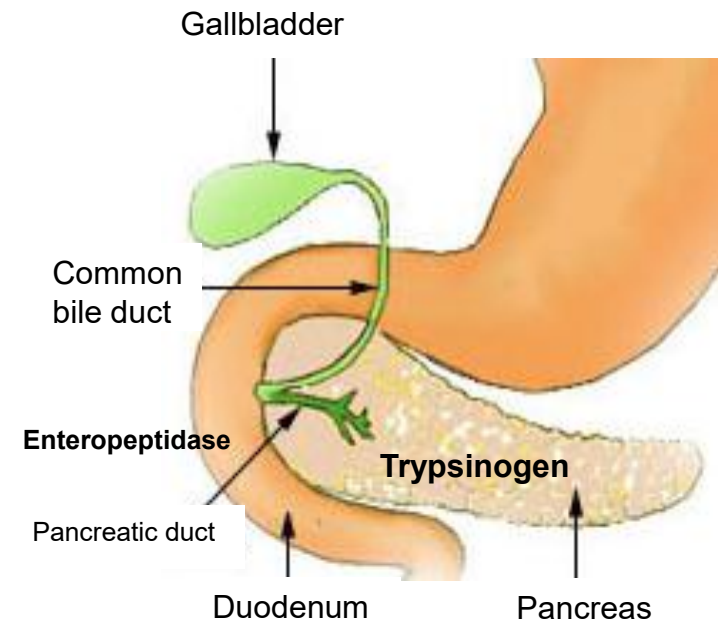
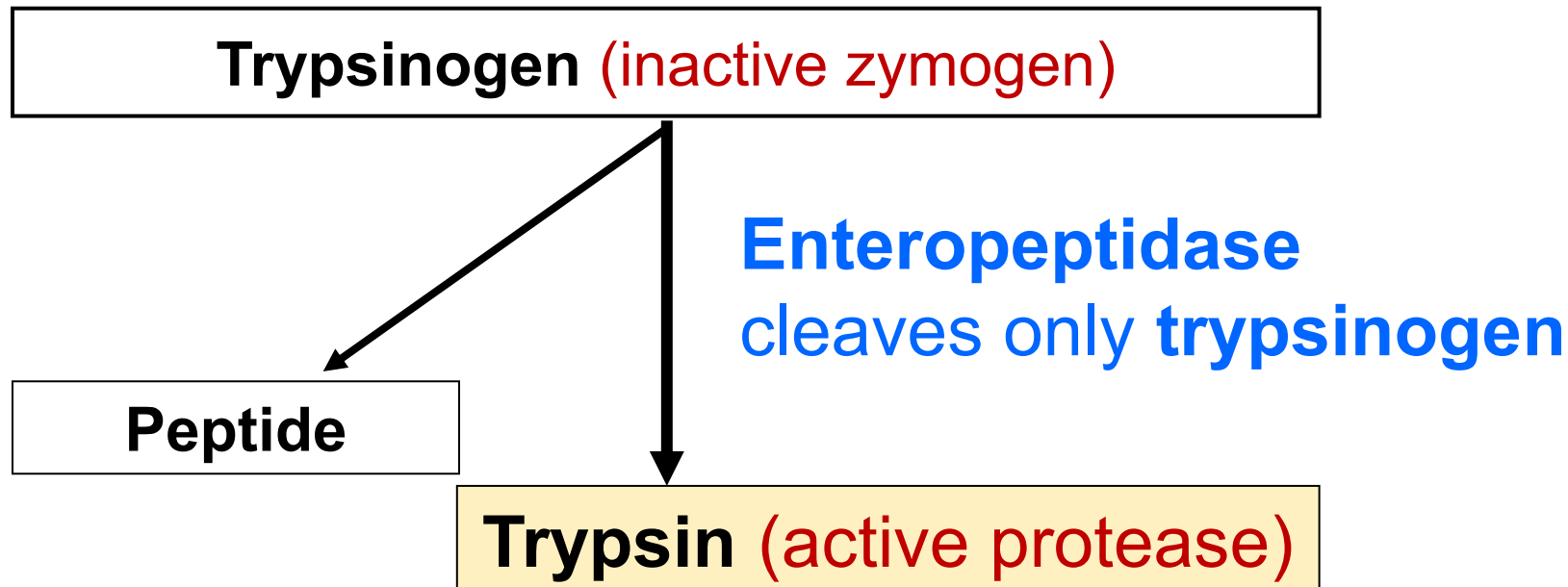


1. **Mouth:** proteins are not degraded.
2. **Stomach:** **pepsin** degrades proteins to **polypeptides** and **amino acids**
Note: **low pH is needed (see DLA)**.
3. **Small intestine:** **pancreatic proteases** form **oligopeptides** and **amino acids**.
Note: **neutral pH is needed**
4. **Brush border cells:** **peptidases** form **amino acids** which are absorbed in the small intestine and reach the liver via the hepatic portal vein.



Protein Digestion in the Small Intestine

- ★ **Activation of most zymogens (proenzymes) occurs in the intestinal lumen**
- **Trypsinogen** is released by the **pancreas** and is **cleaved to trypsin only after it reaches** the **lumen** of the **duodenum** by the activated **enteropeptidase**.
- This **local separation** prevents **pancreatic damage** as **trypsin** is a powerful **protease** and **activates** all other **pancreatic zymogens**.



Proteolytic Activation of Pancreatic Zymogens.

Once **trypsinogen** is activated by enteropeptidase, **trypsin** activates the following **zymogens** for:

1. Protein digestion:

- **Trypsinogen** to **trypsin**
- **Chymotrypsinogen** to **chymotrypsin**
- **Proelastase** to **elastase**
- **Procarboxypeptidases** to **carboxypeptidases**

2. Lipid digestion:

- **Prophospholipase A₂** to **phospholipase A₂**
- **Procolipase** to **colipase (protein)**

Pancreatic proteases simultaneously cleave proteins after specific amino acid residues

- **Trypsin:** after Arg or Lys residues.
- **Chymotrypsin:** after bulky and aromatic residues.
- **Elastase:** after Gly, Ala, Ser residues.
- **Carboxypeptidases A or B** (exo-peptidases) cleave amino acids from the carboxyl-end.

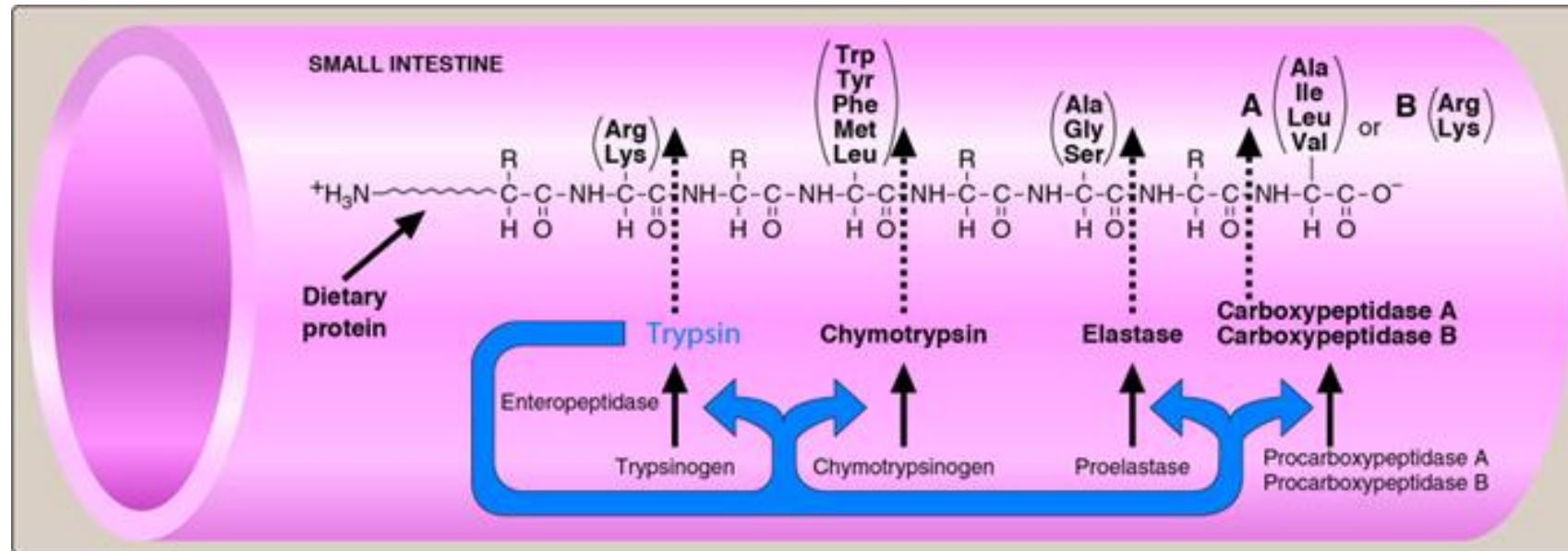
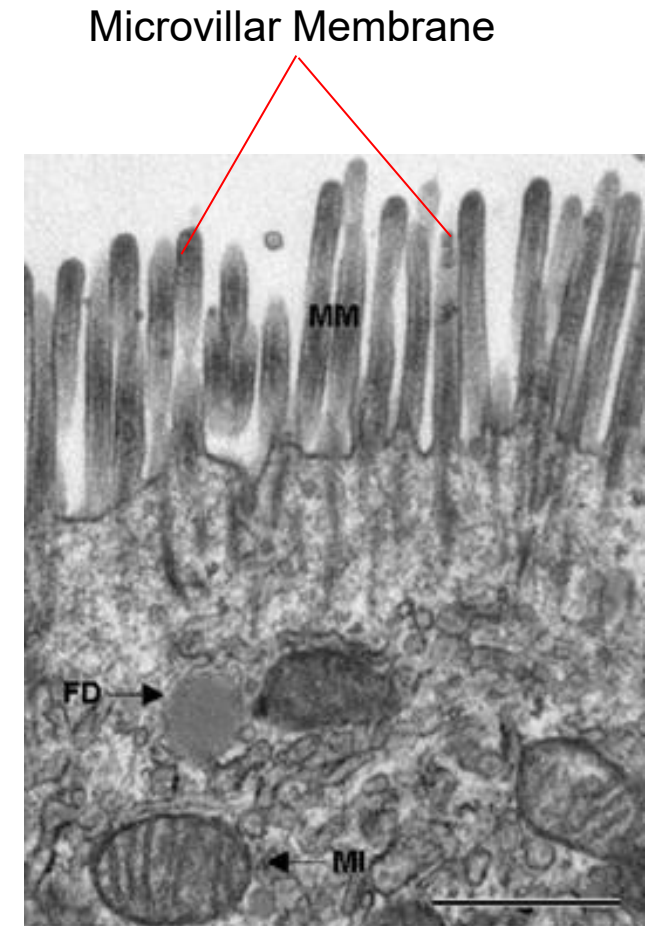


Figure 19.5 Lippincott® Illustrated Reviews: Biochemistry, 9e

- After the dietary proteins are cleaved, the digestive proteases cleave other digestive enzymes and each other.

Brush Border Peptidases

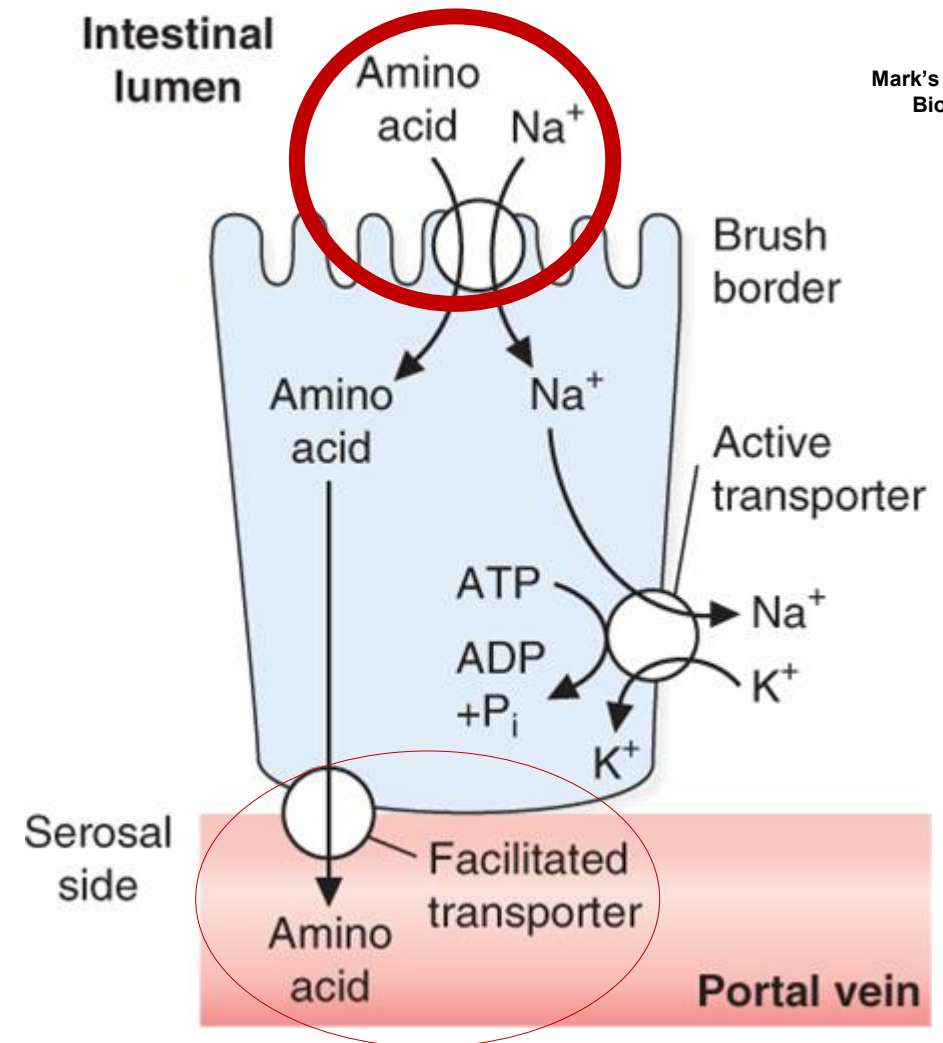
- **Peptidases:**
 - Aminopeptidases
 - Carboxypeptidases
 - Di- and Tripeptidases
- **Convert poly- and oligopeptides to their monomers (amino acids)**



Danielson & Hanson. BBA Biomembranes. 2003

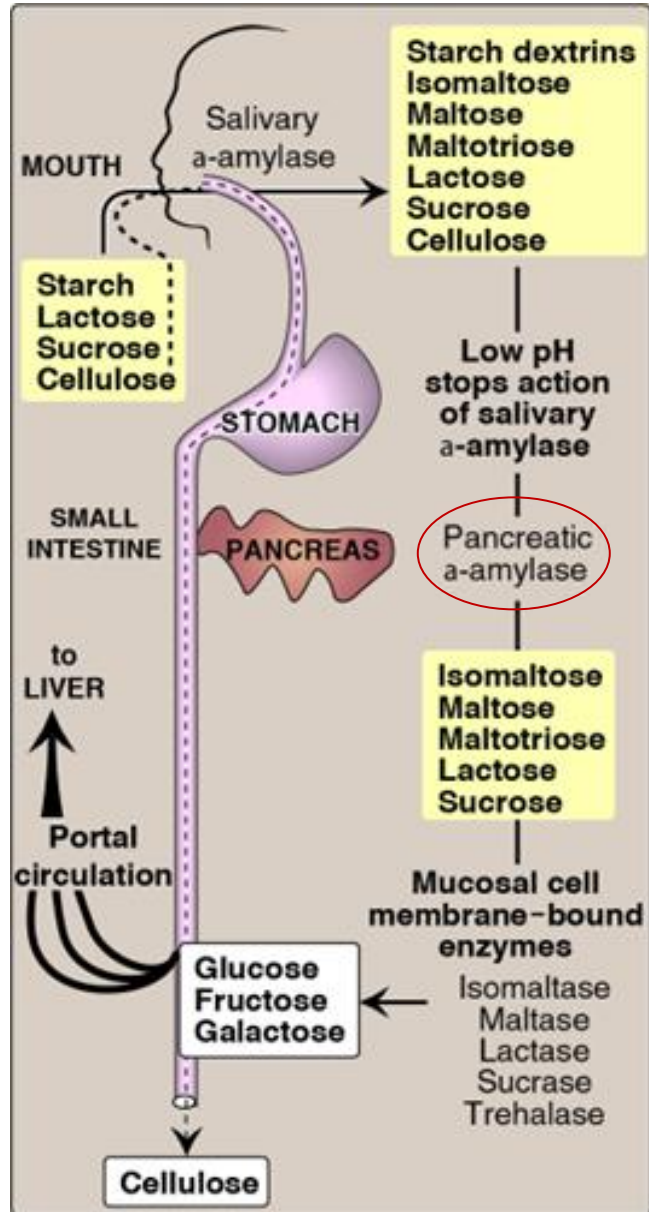
Absorption of Dietary Amino Acids

- The **absorption** of dietary amino acids is mainly performed by **secondary active transport** with **cotransport** of **sodium ions**.
- **Several semi-specific transport systems exist for amino acids (AAs).** The transporters can transport a range of amino acids across cell membranes, though they may have a preference for certain types of AAs.
- The entry into the **hepatic portal vein** is by facilitated transport.
- **Hartnup disease:** Inherited disorder due to a defect in transporter for **Tryptophan (neutral amino acid transporter)**. Tryptophan is also lost in urine due to decreased reabsorption





BIG picture: Carbohydrate Digestion



1. **Mouth:** salivary α -amylase degrades **starch** and **glycogen** to dextrins, isomaltose and maltose.
Note: Humans cannot digest cellulose (lost in feces).

2. **Stomach:** carbohydrate digestion **stops due to acidic pH** and salivary amylase is digested by **pepsin**.

3. **Small intestine:** pancreatic α -amylase forms limit dextrin, maltotriose, isomaltose and maltose..

4. **Intestinal Brush border cells: Disaccharidases** cleave
a. **Lactose** to glucose and galactose and
b. **Sucrose** to glucose and fructose
c. **Maltose** and **isomaltose** to glucose

The monosaccharides are absorbed in the small intestine and reach the liver via the hepatic portal vein.

Absorption of Dietary Monosaccharides

- Dietary **glucose** and **galactose** enter the intestinal mucosal cells via **SGLT-1** which uses **secondary active transport** and **cotransport** with **sodium ions**.
- Dietary **fructose** enters the intestinal mucosal cells via **facilitated transport** by **GLUT-5**.
- All three monosaccharides enter by **facilitated transport** via **GLUT-2** into the **hepatic portal vein**.

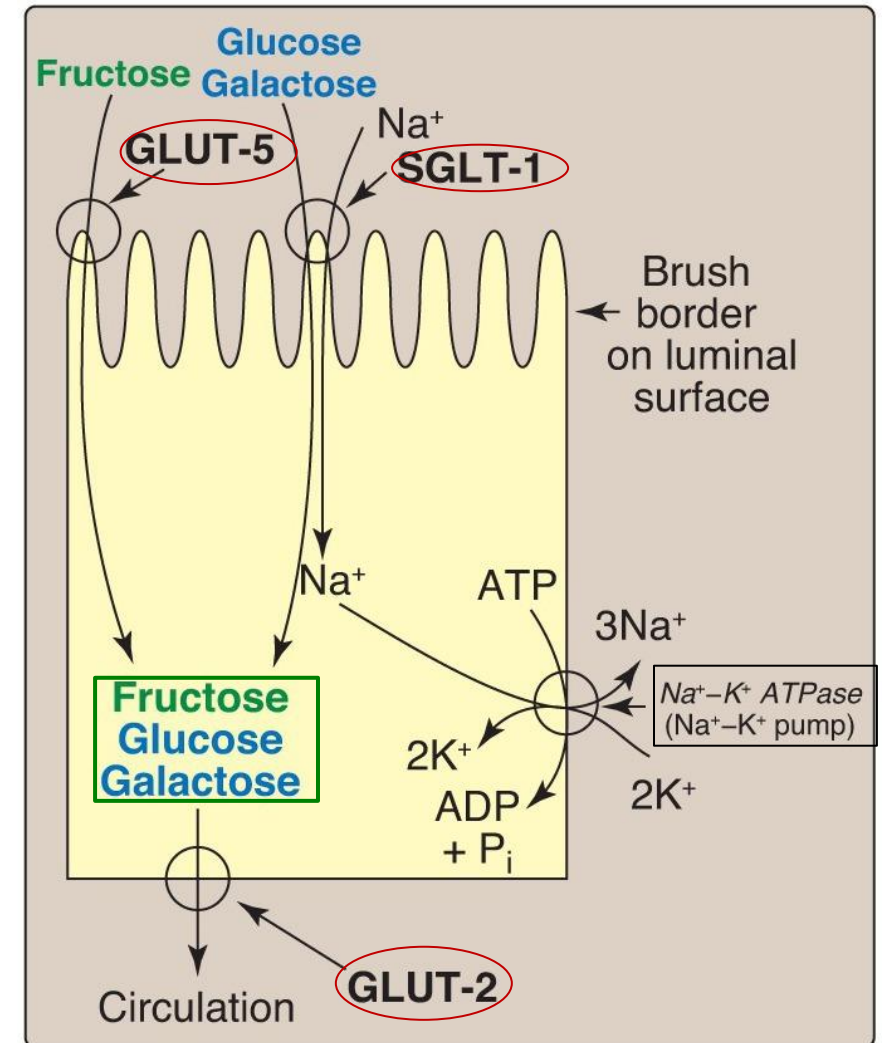


Figure 7.10 Lippincott® Illustrated Reviews: Biochemistry, 9e



BIG picture: Lipid Digestion

- 1. Mouth:** lingual lipase is mainly swallowed.
- 2. Stomach:** lingual and gastric lipase degrade **milk TAGs** with **medium-chain FA** at acidic pH.
- 3. Small intestine:** Conjugated bile salts and peristalsis are needed for **TAG digestion** by **pancreatic lipase**, forming **MAG** and 2 long-chain fatty acids
- 4. Brush border cells:** absorption of free **fatty acids**, **MAG**, free **cholesterol**, lysophospholipids, and lipid-soluble **vitamins** needs **mixed micelles** with **conjugated bile salts**.
- 5. Chylomicrons** are formed in intestinal cells and **released into the lymph**.

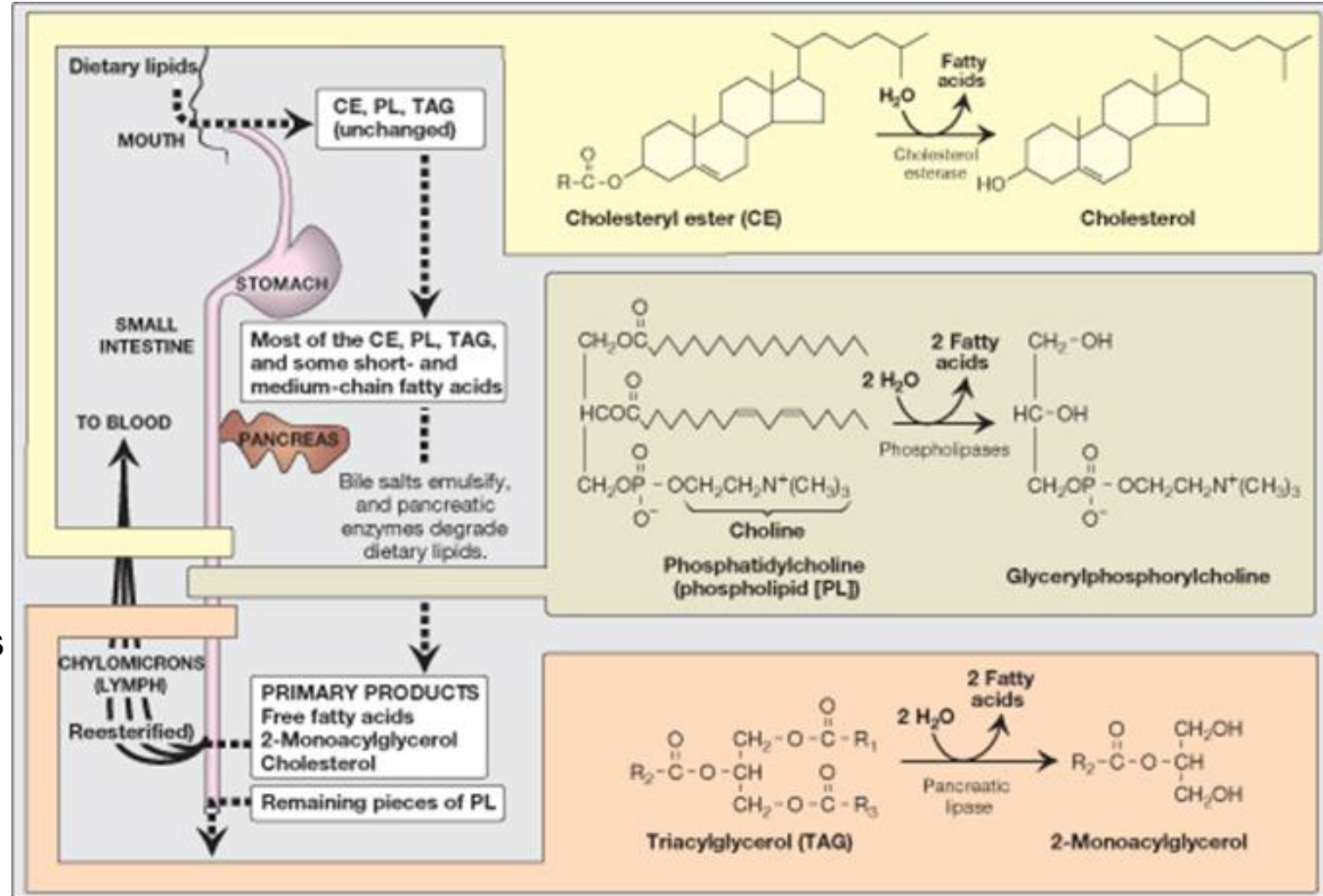
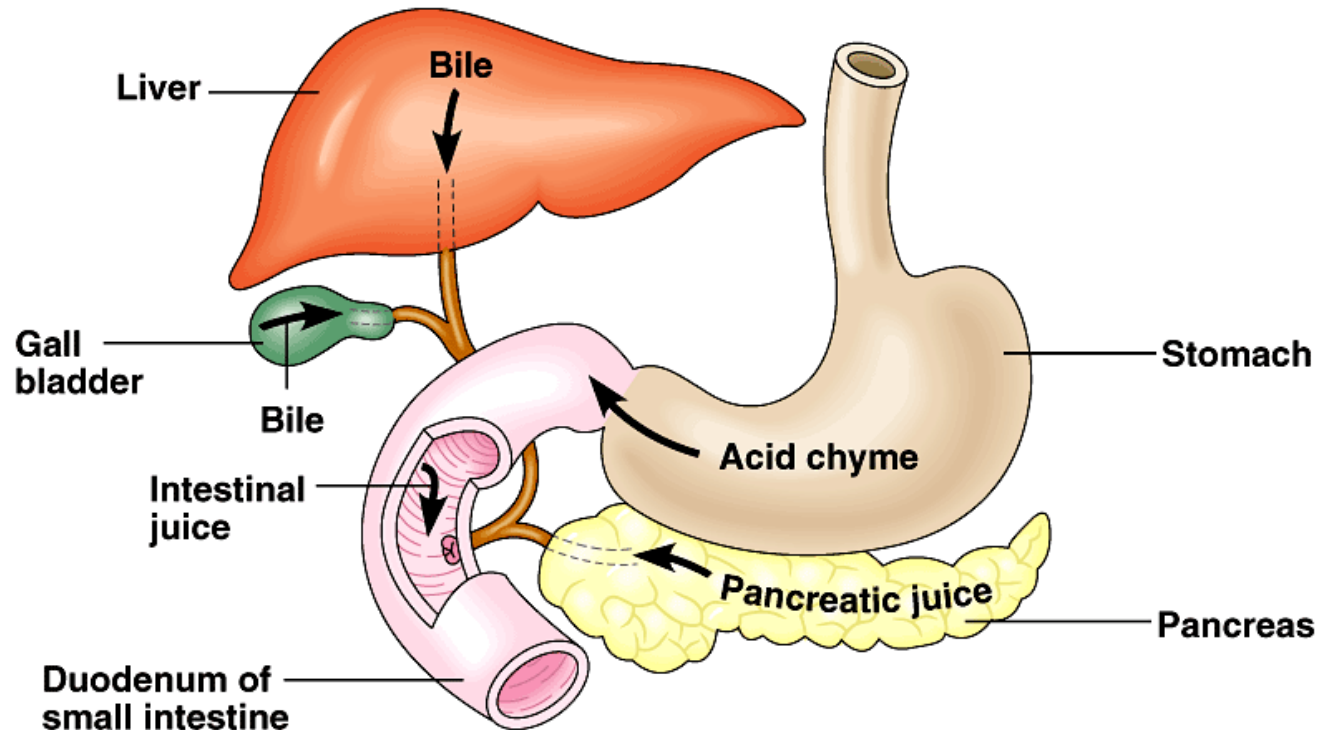


Figure 15.2 Lippincott® Illustrated Reviews: Biochemistry, 9e

Lipid Digestion and Absorption in Small Intestine Needs Conjugated Bile Salts

Bile is an **alkaline fluid** formed in the liver and contains **bile salts**, cholesterol, bilirubin, and other substances.

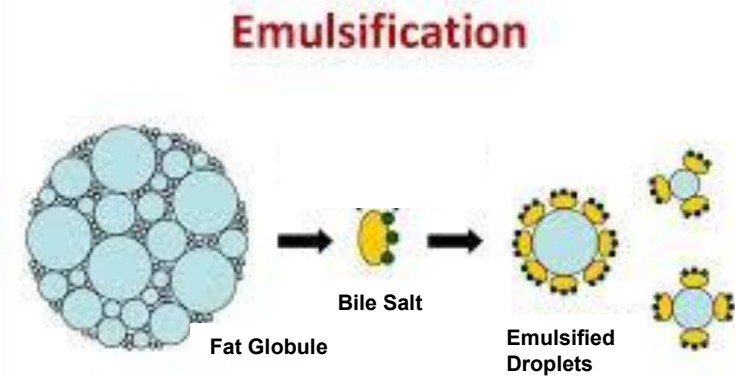
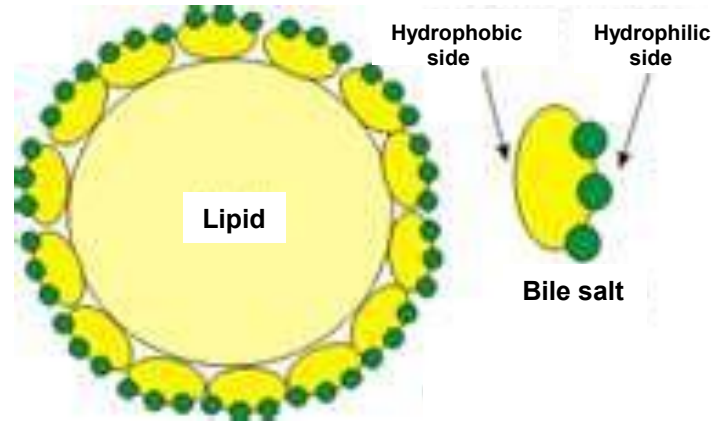
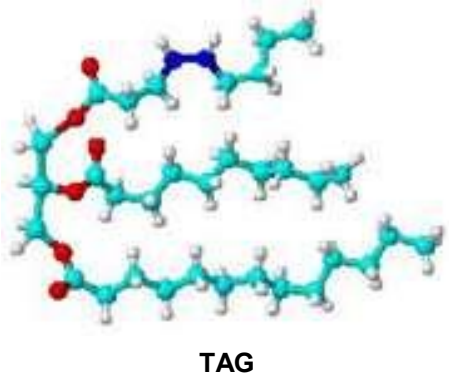


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1. **Conjugated bile salts emulsify** the **dietary TAGs** for digestion by pancreatic lipase
2. **Absorption** of MAGs, cholesterol and free fatty acids needs **mixed micelles** formed with conjugated bile salts.
3. **Blockage** of bile entering the duodenum by cholesterol gallstones leads to dietary fat maldigestion and malabsorption

Functions of Conjugated Bile Salts

Emulsification of dietary lipids in the small intestine for digestion by pancreatic lipase and cholesteryl esterase and phospholipase A2.

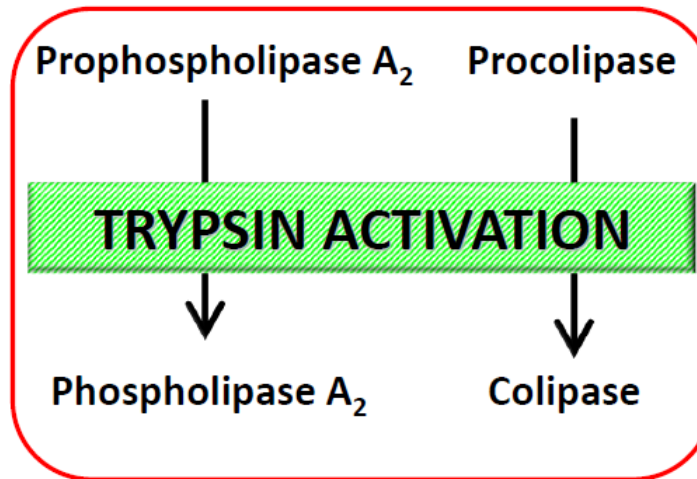


- Emulsification increases the surface area of the hydrophobic lipid droplets so that the digestive enzymes can act effectively.
- Emulsification is accomplished by the detergent properties of the conjugated bile salts and mechanical mixing due to peristalsis.

Enzymes of Lipid Digestion

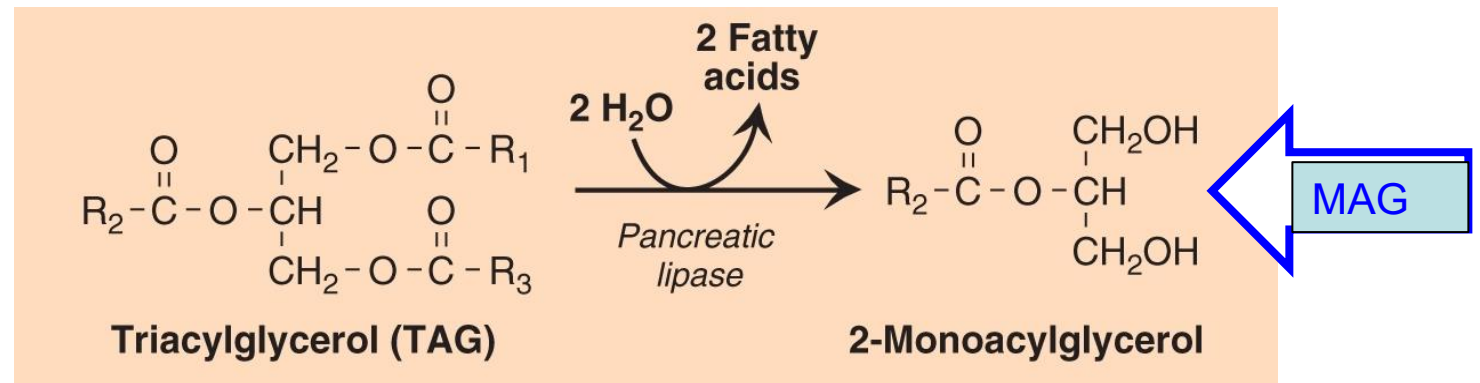
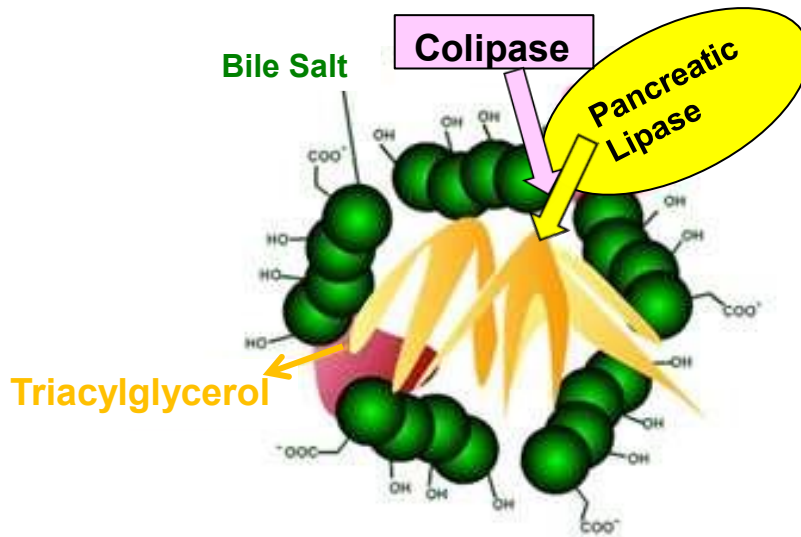
- Cholesterol Esterase
 - Phospholipase A₂
- } **Activity increased**
in the presence of **bile salts**

- ★ • **Pancreatic lipase** requires **Colipase** (a protein) for optimum activity.
- **Proteolytic activation** required for some enzymes:



Pancreatic lipase cleaves TAG to MAG and 2 fatty acids

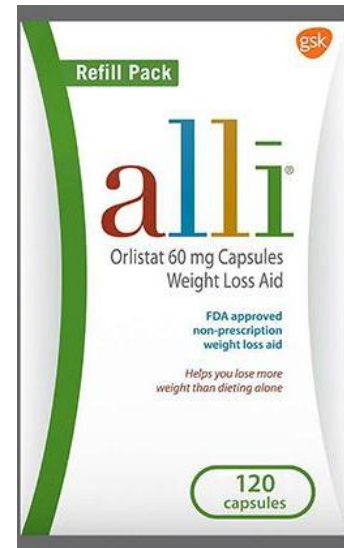
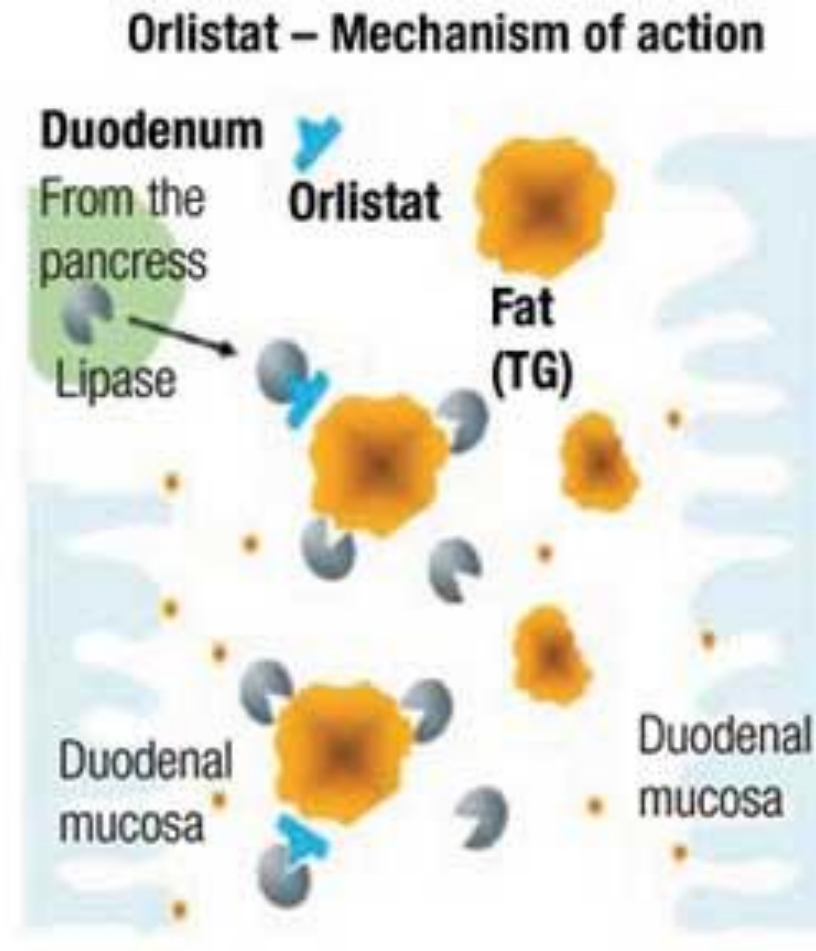
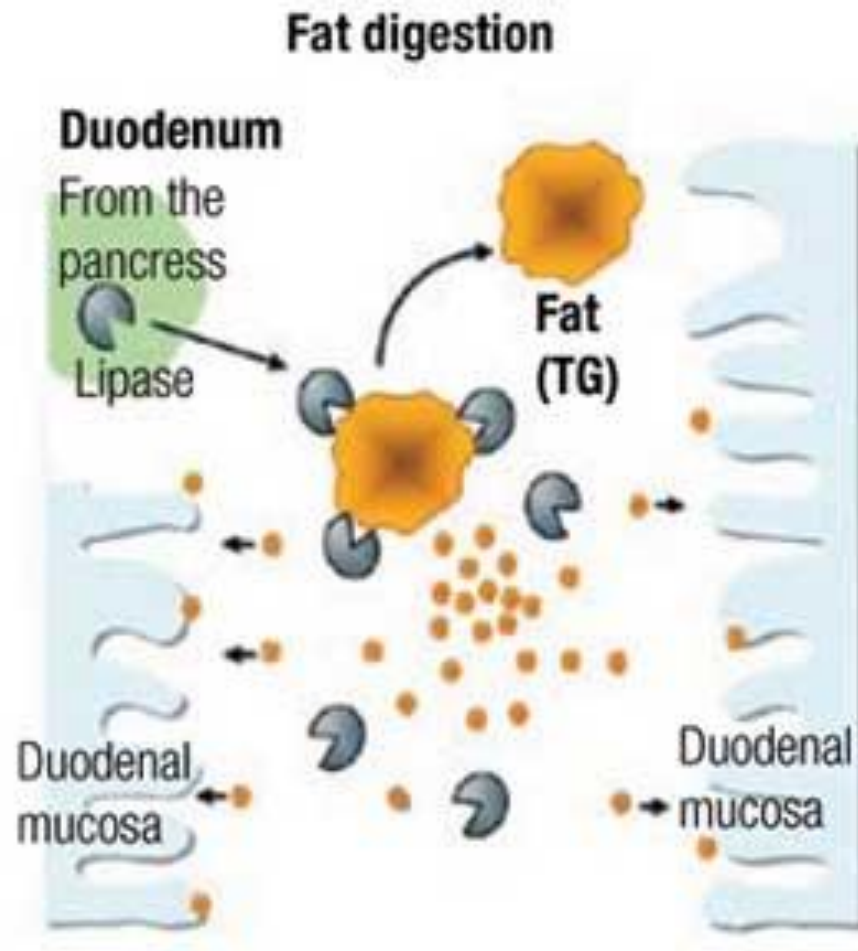
- The protein **colipase** anchors **pancreatic lipase** to the **lipid droplet**.
- **Pancreatic lipase** converts TAG to Monoacylglycerol (MAG) that can enter the intestinal mucosal cells via micelles
- End product of TAG digestion is **MAG and fatty acids**



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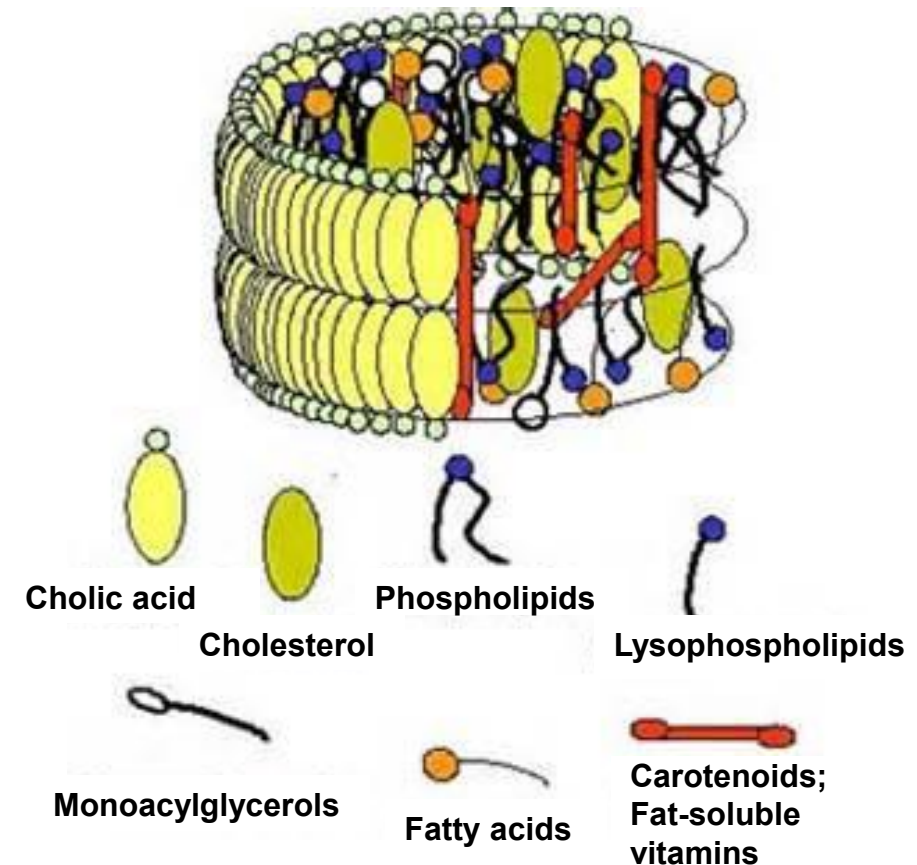
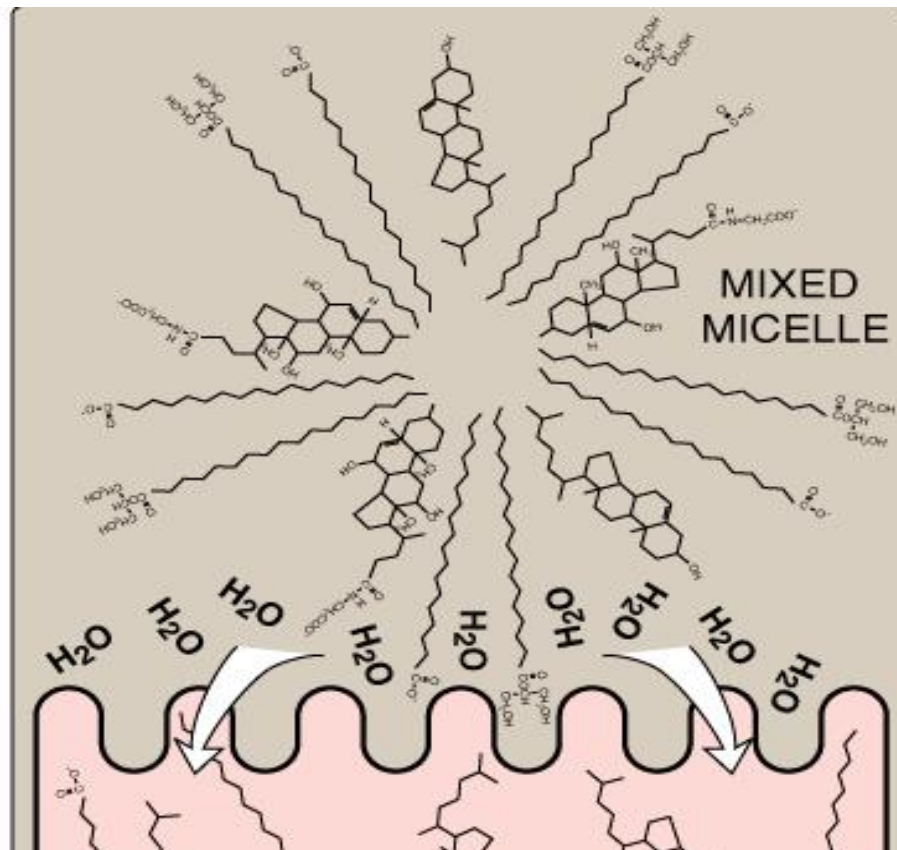
Clinical application

Pancreatic lipase inhibited by the drug **Orlistat** (promotes weight loss)



Functions of Conjugated Bile Salts

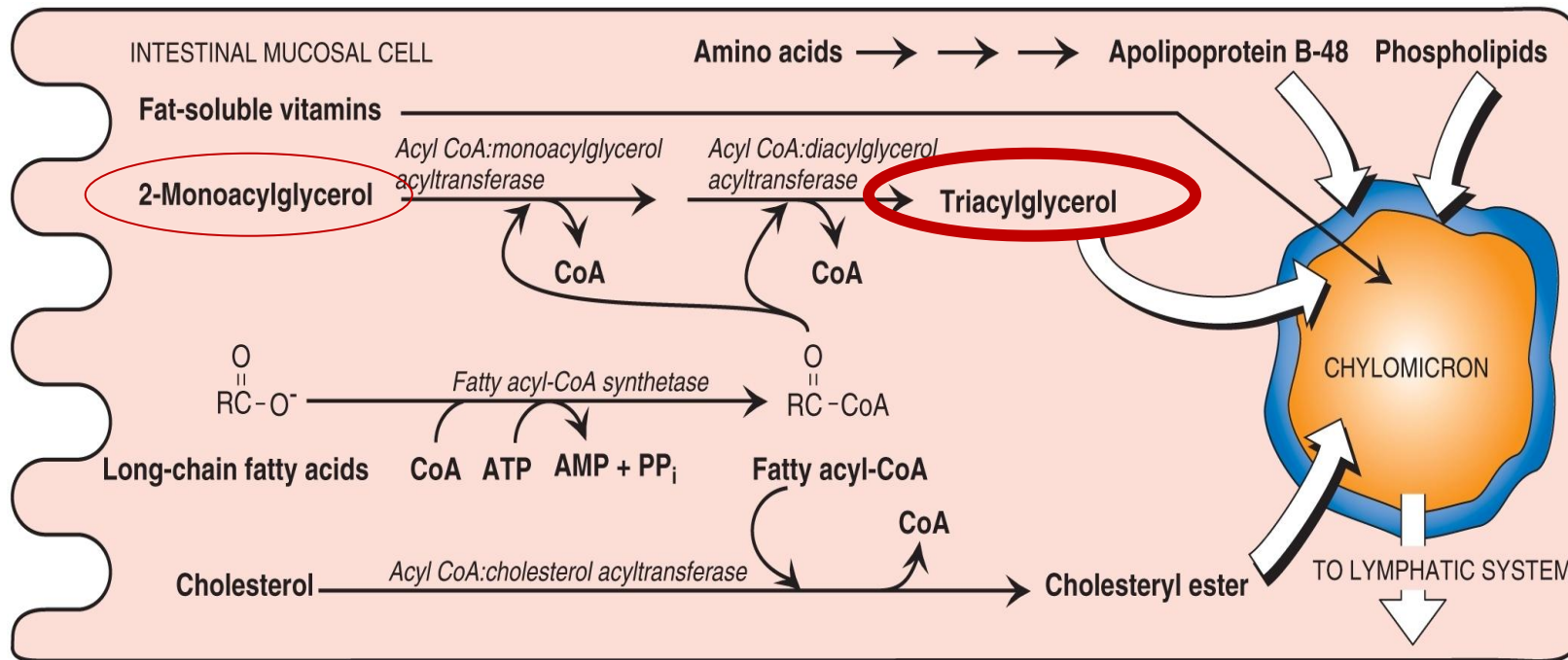
Facilitates absorption of digested lipids into intestinal mucosal cells by forming mixed micelles.



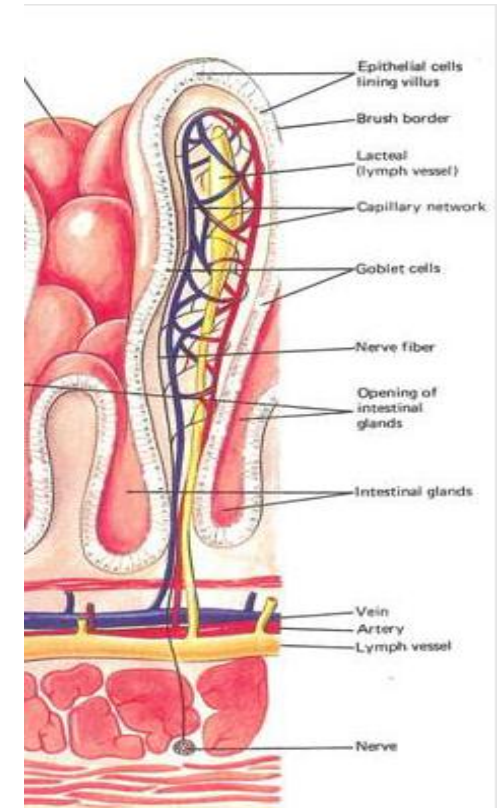
SCFAs, MCFAs, and glycerol directly enter portal blood (not lymphatics) – Do **NOT** require micelles

Synthesis of Chylomicrons: Following Absorption of Dietary Lipids:

- Intestinal mucosal cells resynthesize TAG from **MAG**
- The **MAG pathway** uses absorbed MAG and fatty acids to synthesize TAG



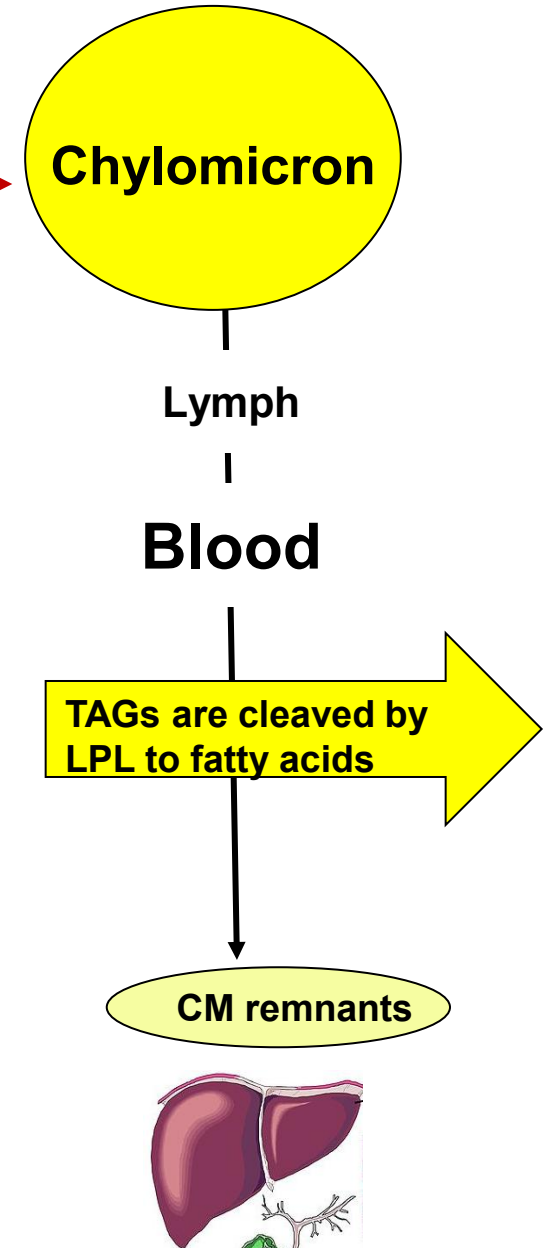
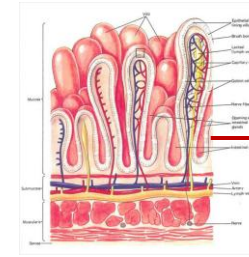
Note: Special TAG synthesis in intestinal mucosal cells



SOM. MK.I.BPM2.3.DM.3.BCHM.1250

Review: Fate of Chylomicrons

- **Chylomicrons** filled with **dietary lipids** are released into the **lymph** and reach the **blood**.
- The **dietary TAGs** in **chylomicrons** are cleaved by **lipoprotein lipases (LPL)** in **capillaries** of the **heart** and **fat cells**.
- The formed **fatty acids** are used for **energy metabolism** in **heart** and **muscle** or are **stored** in **fat cells** as **TAGs**. (LPL activity facilitates lipid storage in adipocytes)
- The **chylomicron remnants** containing **cholesteryl ester**, and **lipid-soluble vitamins** are **taken up** by **liver** by **endocytosis**.



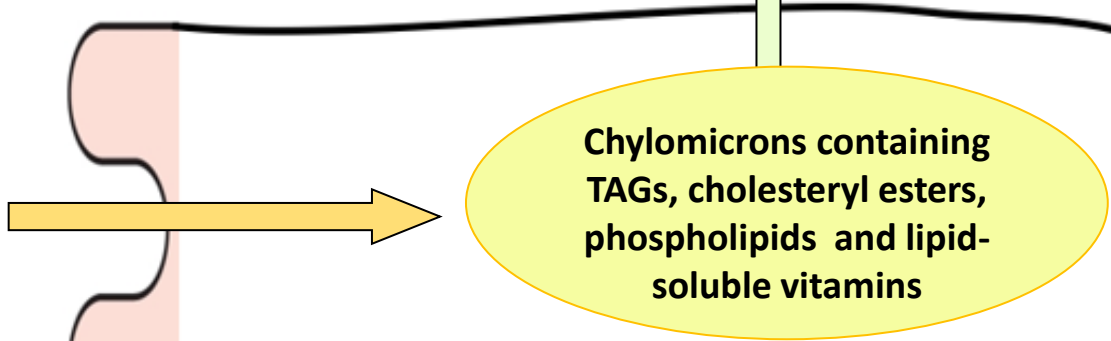
Destruction of villi/ mucosal cells affects nutrient absorption

Summary of Absorption

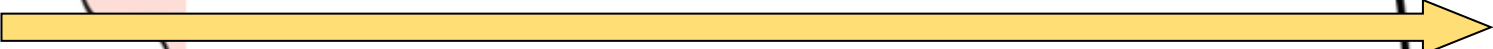
Lymph

Intestinal lumen

Cholesterol, 2-MAG
LCFAs, Lysophospholipids
Fat-soluble Vitamins



SCFAs, MCFAs, Glycerol



Amino acids

Na⁺ co-transport



Amino acid transporter



Fructose

GLUT-5

Glucose or Galactose

SGLT-1

Na⁺ co-transport



GLUT-2



DNA/ RNA



Purines



Uric acid

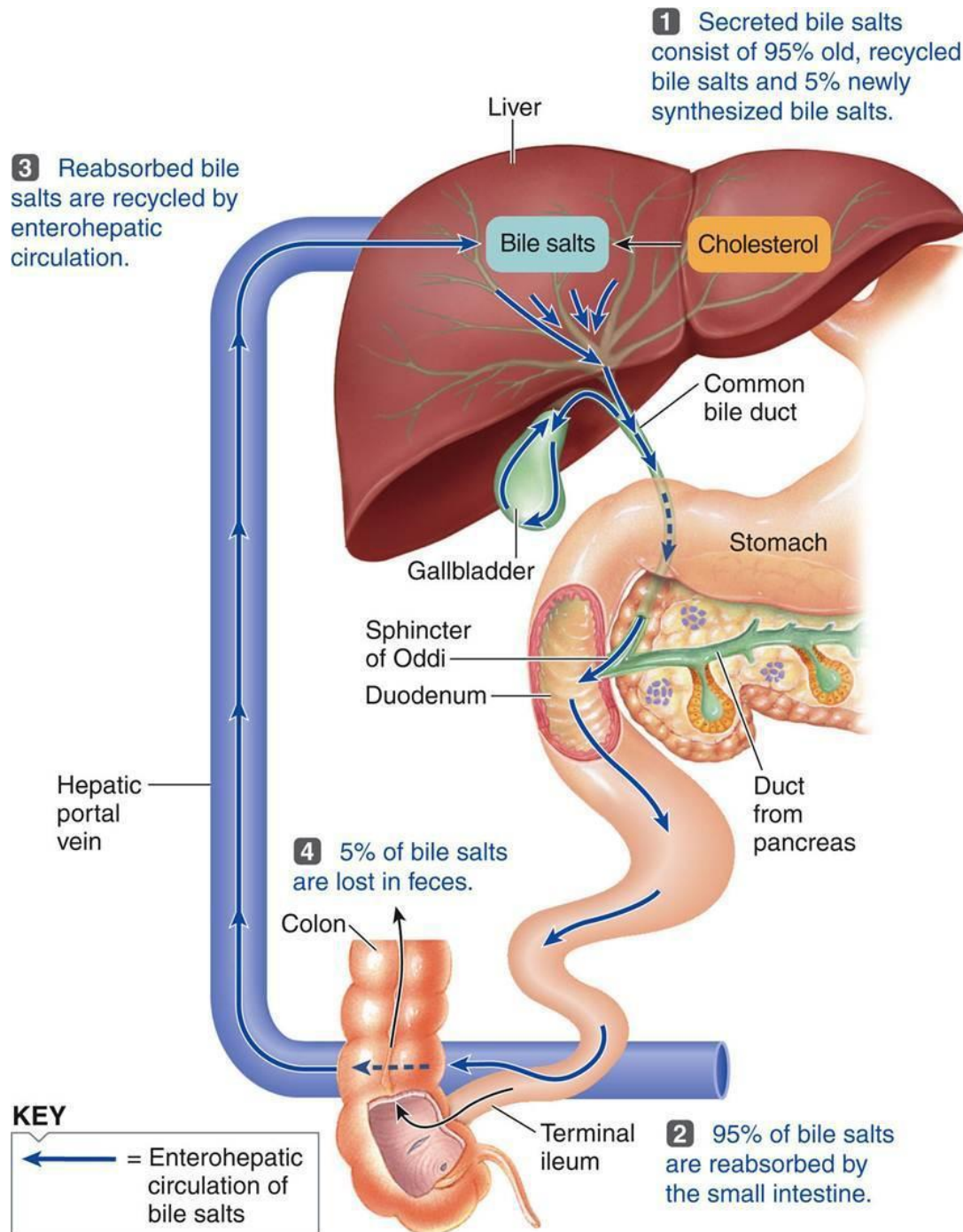
Urine
via blood



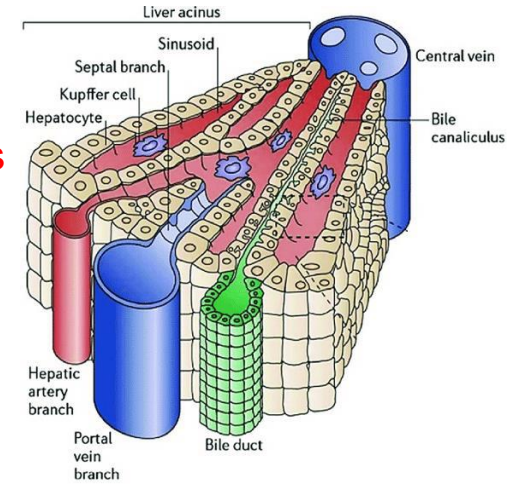
Portal vein
To Liver

ENTEROCYTE

Formation and Enterohepatic Circulation of Bile Salts/Bile Acids



Hepatocytes release bile salts into bile canaliculi → gall bladder



1. Formation of conjugated (primary) bile salts in liver and release into duodenum via bile ducts
2. Reabsorption of 95% of the primary bile salts/acids from **terminal ileum**
 - Formation of secondary bile acids by bacteria
3. Transport of primary bile acids back into the liver via the hepatic portal vein (95% recycled) – Conjugated and re-secreted into bile (Enterohepatic circulation)
4. Loss of bile salts in feces (5% lost in feces) - Secondary bile acids

Cholelithiasis

- Presence of stones in the gallbladder
- **Cholesterol** stones (~80%) of gallstones.
- **Causes of Cholesterol Gallstone Disease:**
 - **Imbalance in ratio of bile salts: cholesterol in bile** (leads to **poor** cholesterol solubility)
 - **Reduction** of hepatic **conjugated bile salt** secretion
 - Increased **biliary cholesterol** secretion
 - Gallbladder hypomobility

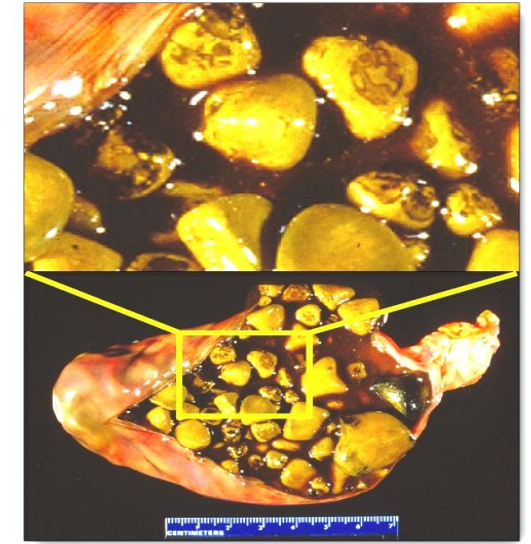
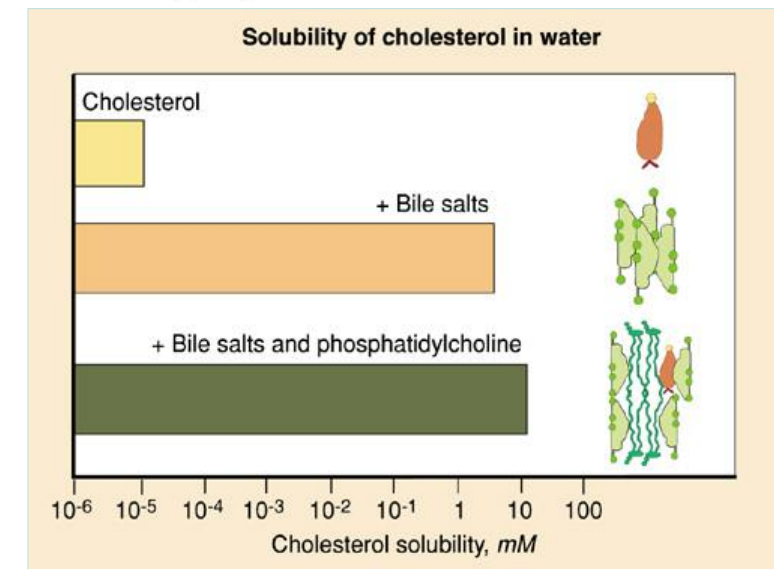
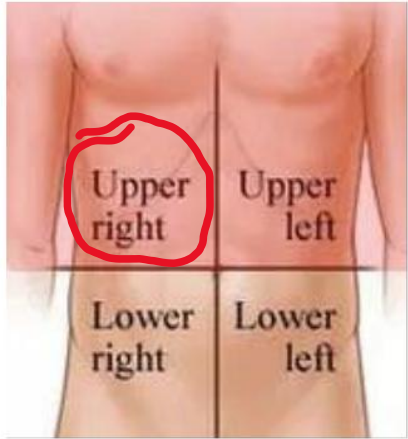


Figure 18.12
Gallbladder with gallstones.

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Gallstones effects: Biliary Colic, Maldigestion & Malabsorption of Lipids



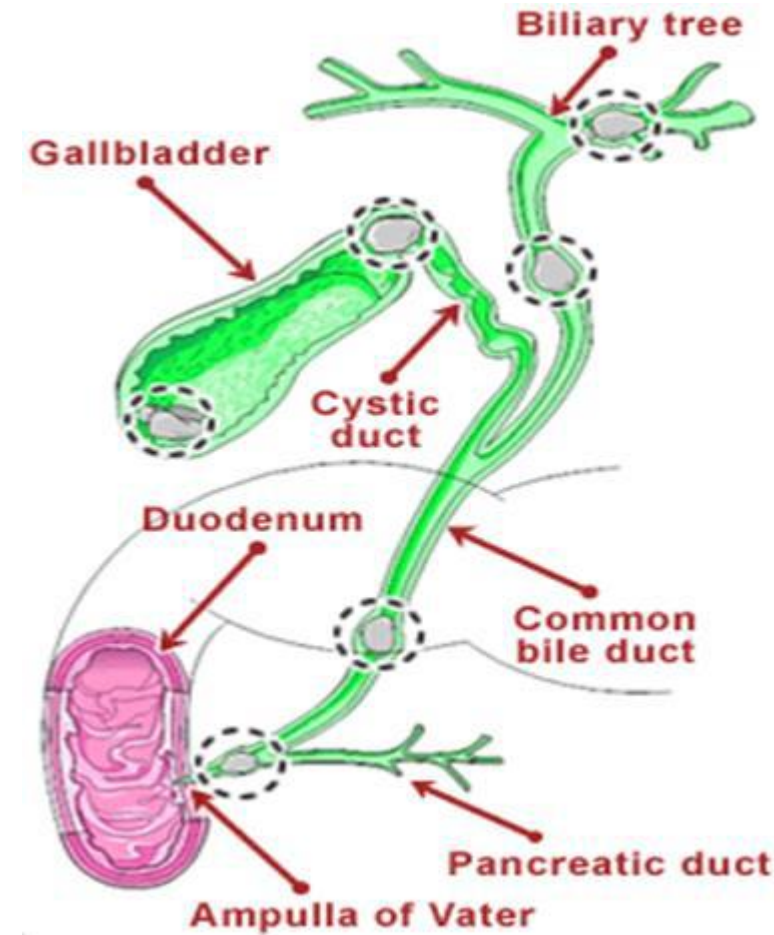
- **Biliary Colic** - most common complication
- Stone blocks cystic duct or common bile duct
- Severe right upper abdominal pain that may radiate to the shoulder

Bile duct obstruction impairs dietary **lipid digestion** as **bile salts** do not reach small intestine resulting in:

- Less emulsification of lipids.
- Less uptake of digested lipids.

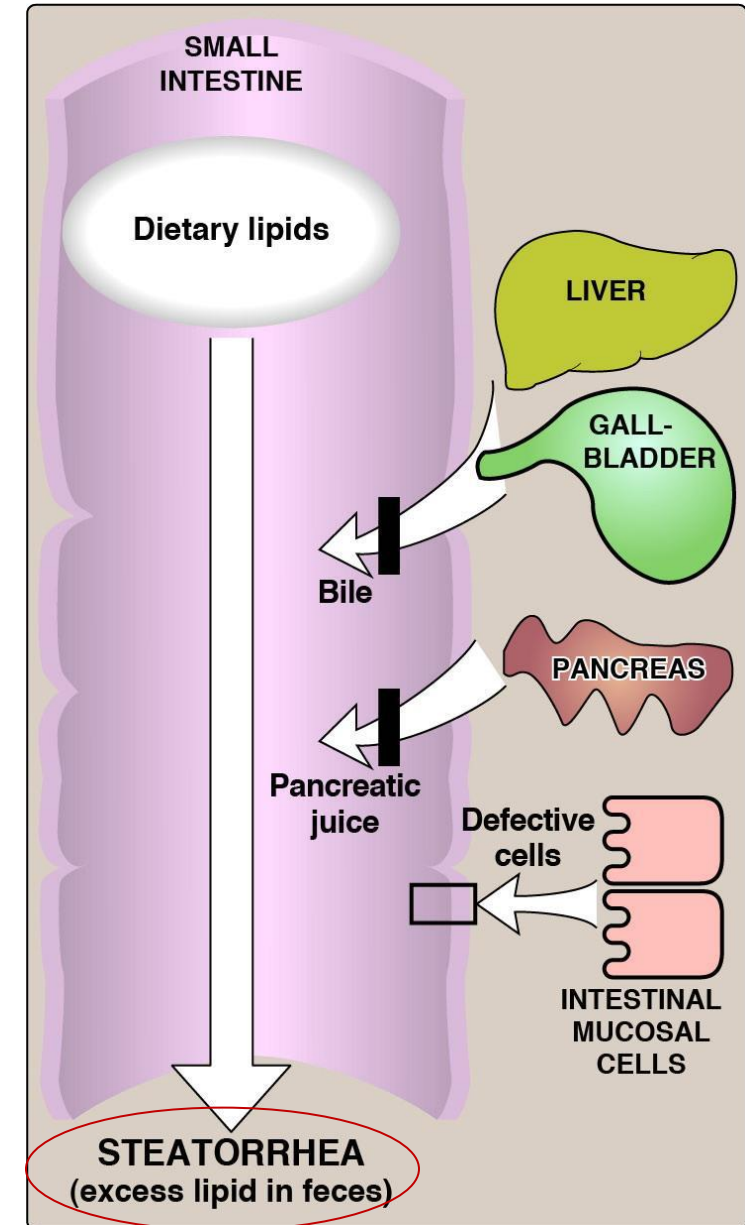


Steatorrhea



Causes of Steatorrhea

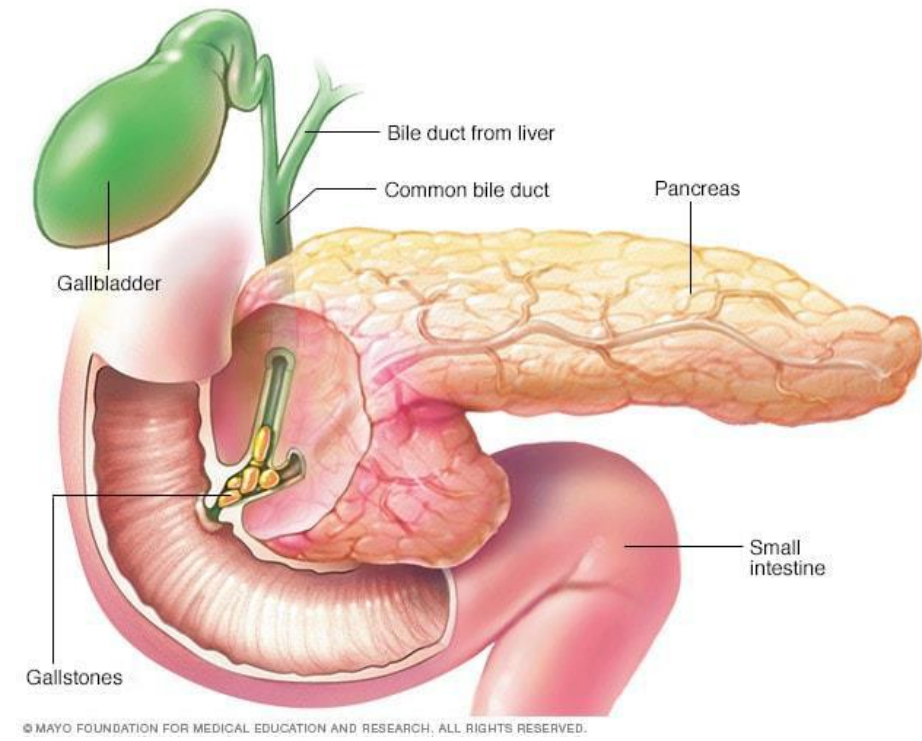
- **Excessive fat in the stools**
- **Lack of bile salts due to:** (Impairs lipid digestion and absorption/ micelle formation)
 - **Liver damage** including liver **cirrhosis**.
 - **Bile duct obstruction by stones** resulting from an imbalance of cholesterol and conjugated bile salts in bile.
 - Bile duct obstruction by **pancreatic tumor**
- **Defects related to pancreatic juice:** Deficiency of enzymes: Impairs lipid digestion
 - **Lack of transport** into the intestines in **cystic fibrosis (CF)**
 - **Lack of bicarbonate secretion** (Impairs pH in the duodenum)
- **Defective intestinal mucosal cells or shortened bowel** (Impairs absorption and chylomicron formation)



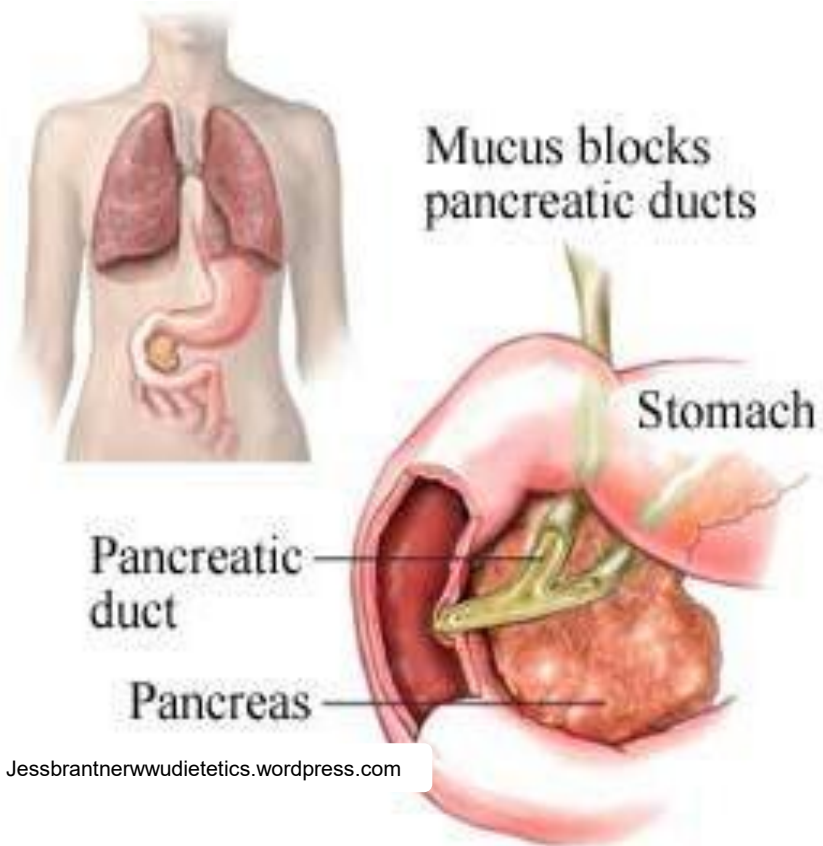
Pancreatic Diseases:

Stones can lead to Pancreatitis

- **Stones** can **block** pancreatic duct preventing **zymogen** release into **duodenum** causing **pancreatitis**.
- **Trypsin** is **pre-maturely active** in pancreas. **Activates** pancreatic **zymogens** which destroy proteins and phospholipid membranes resulting in pancreatitis.
- **Pancreatic enzymes** are not secreted into **duodenum** which **impairs digestion of all food groups** (Lipids/ proteins and carbohydrates)



Pancreatic Insufficiency in Cystic Fibrosis



CF patients often have chronic pancreatitis, steatorrhea and are undernourished.

- Defective CF transmembrane conductance regulator (CFTR) leads to decreased secretion of chloride and increased uptake of sodium and water in exocrine glands
- **In the pancreas, depletion of water results in thickened mucus that clogs pancreatic ducts. Mucus plugs prevent pancreatic enzymes from reaching intestine, thereby leading to pancreatic insufficiency.**
- **Less pancreatic zymogens and enzymes reach the duodenum. Digestion of proteins and lipids is mainly reduced**
- Pancreatic amylase is efficient at low enzyme concentration

Cystic Fibrosis Dietary Therapy

- **Goal – maintain adequate nutrition**
- Diet rich in carbohydrates and protein (allows growth)
- Low fat diet to prevent steatorrhea
- **TAGs with medium-chain fatty acids** may be added, as this leads to **lipid digestion in the stomach** without the need for pancreatic lipase. The medium-chain fatty acids are taken up into the portal vein and directly reach the liver for energy metabolism.
- Vitamin supplements
- Supplement with porcine digestive enzymes



Summary of different Lipases

Lipases	Synthesized in	Active in	Purpose
Lingual lipase Gastric lipase	Salivary gland cells Gastric chief cells	Stomach lumen Acid-stable enzymes Need low pH	Digestion of TAGs with medium-chain fatty acids (milk). Important for babies and patients with CF. No bile salts needed.
Pancreatic lipase	Acinar cells of exocrine pancreas	Intestinal lumen which needs bile salts for emulsification of dietary TAGs and colipase for anchoring to TAGs	Cleaves TAGs to MAGs for absorption into intestinal mucosal cells. Provides MAG for TAG synthesis in intestinal mucosal cells.
Lipoprotein lipase (LPL)	Myocytes, adipocytes	Blood: LPL is anchored in capillaries of heart, muscle and fat tissue. Activated by insulin and Apo CII	Cleaves TAGs in chylomicrons and VLDLs and provides free fatty acids for uptake into heart and muscle (energy metabolism) or into fat cells for synthesis and storage of TAGs.
Hormone sensitive lipase	Adipocytes	Adipocytes Activated by low insulin: glucagon and epinephrine	Degradation of TAGs stored in fat cells (lipolysis) and release of free fatty acids and glycerol into blood during fasting and fight or flight response

Summary: Functions of conjugated bile salts

- **Emulsification of dietary lipids** in the small intestine needed for **pancreatic lipase, cholesteryl esterase** and **PLA2** activity
- **Formation of mixed micelles** for **dietary lipid absorption** into intestinal mucosal cells.
- **Use of free cholesterol** in hepatocytes for **synthesis of primary bile acids**
- **Transport of free cholesterol** from liver into **duodenum** (loss of **free cholesterol** in **feces**)

Reminder: Only the liver is involved in:

- **Synthesis** of **primary bile acids** using **free cholesterol**.
- **Conjugation** of primary and secondary bile **acids** with **glycine** or **taurine** to form **bile salts**.
- **Release of bile salts** and **free cholesterol** into **bile** for transport into **duodenum** to **assist dietary lipid digestion** and **absorption**.
- **Loss of cholesterol** in **feces** (directly by free cholesterol and indirectly as bile acids)

Summary: Carbohydrate digestion in the mouth and small intestine

Salivary amylase cleaves starch and glycogen to small branched oligosaccharides (dextrins).

Note: Disaccharides not digested until they reach the intestinal mucosa cells.

Pancreatic amylase acts on **polysaccharides** and **dextrins** to generate **maltose** and isomaltose **which contain only glucose**.

Isomaltase and maltase (Brush border disaccharidases) form **glucose** from **maltose** and isomaltose.

Dietary disaccharides (Are **not** degraded by salivary or pancreatic amylase):

- **Sucrase** degrades **sucrose** to **glucose** and **fructose**.
- **Lactase** degrades **lactose** to **glucose** and **galactose**.

Summary: Lactose Intolerance (Disaccharidase deficiency)

1. Primary lactose intolerance is also known as **lactase non-persistence** or **adult hypolactasia**.

- Due to **age-dependent decline** of **lactase synthesis** in many individuals.
- Lowest levels often at about 7 years of age and later

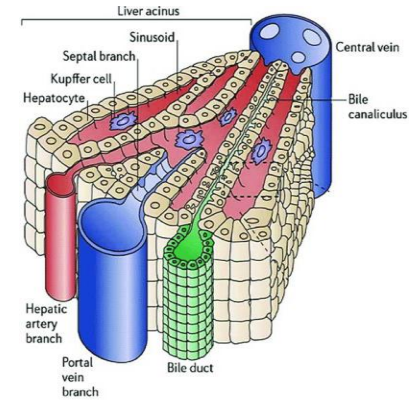
2. Secondary lactose intolerance

- Results from **intestinal mucosal damage due to intestinal disease (diarrhea)**
- Lactase is the first enzyme lost, and the last enzyme reinstated.

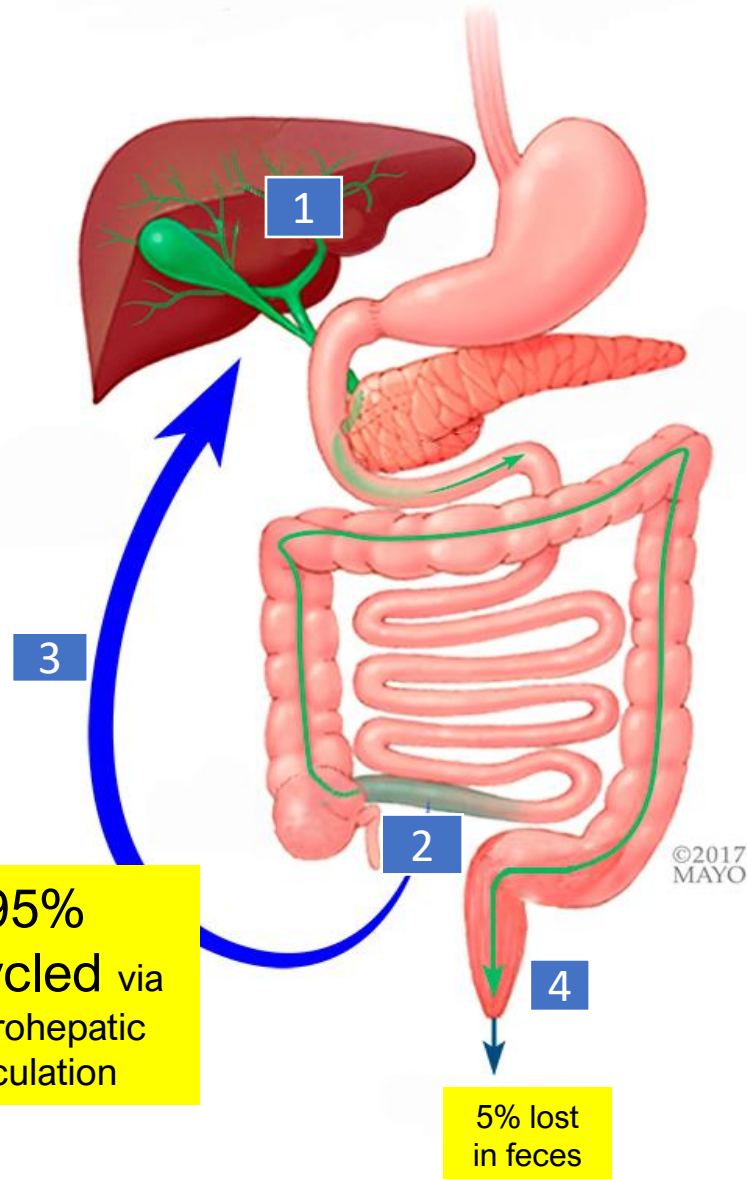
3. Congenital lactase deficiency (most severe form) and rare

- Affects newborns (who normally have the highest activity)
- Severe osmotic diarrhea and dehydration → Death if not treated

Summary: Formation and Enterohepatic Circulation of Bile Salts/Bile Acids



Hepatocytes release
bile salts into bile
canaliculi → gall
bladder



1. Formation of conjugated (primary) bile salts in liver and release into duodenum via bile ducts
2. Reabsorption of 95% of the bile salts/acids from the **terminal ileum**
 - Formation of secondary bile acids by bacteria
3. Transport of primary bile acids back into the liver via the hepatic portal vein (95% recycled) – Conjugated and secreted into bile
4. Loss of bile salts in feces (5% lost in feces) - Secondary bile acids

A large, decorative blue curved shape that starts as a thin line on the left and expands into a solid wedge-like form towards the bottom right corner of the slide.

Thank you.

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